

MS Access to PostgreSQL

Replicator and data sanitiser program

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Overview:

This project has the objective of replicating any MS Access database into the equivalent PostgreSQL database with as little configuration as possible.

In the AI research world, there is almost no development aimed at integrating with user data stored in MS Access databases. There is however significant support for PostgreSQL database, so by creating a replicate of the user data, with the option to update that replica at will, it becomes possible to connect AI platforms to the data normally being managed using MS Access.

This document describes the ‘replicator’ program – not the research tools and processes that might use the replicated data.

The AI open context layer that we want to use the replicated database is [WrenAI](#)

MS Access is a very effective UI and data management tool, but has strong limitations related to capacity and robustness. Microsoft recommends MS SQL Server for users where those limits become a problem. A consequence of that is that as of 2026, Microsoft has marked a number of redistributable support libraries for Access database as ‘no longer supported’. That does not mean MS Access is unsupported, but given that the UI component of Access can use MS SQL Server as its database, and MS SQL Server has an ‘Express’ which has higher capacity limits than MS Access and is free for use, you can deduce that the MS Access database component is being strongly de-emphasized.

I would estimate that many many users of the original MS Access tool have only modest transaction and storage capacity requirements, and very limited budgets for software migrations, so a free tool like this allows them to then use the amazing amount of development in the AI community for analysing data.

This replicator is a bare bones equivalent of MS SQL Server Migration Assistant for a target PostgreSQL database, which is then accessible from many AI integration platforms.

Environments:

The original database that is fed into this replicator is a Microsoft Access database managed via Microsoft Access

The whole concept of 'replicating' a database is not as simple as it seems. New rows should be added, existing rows should be updated, but that only works for tables with some kind of unique key .. tables without keys, can not be updated, only appended to - both in MS Access and PostgreSQL ... so the replicator needs to deal with these subtleties.

The target PostgreSQL database can be created in a Windows, MacOS or Linux system. But the examples in this document show everything operating in a Windows system.

In reality, since the AI systems almost all work in MacOS or Linux systems, the PostgreSQL server would normally operate in one of those environments.

Component software needed

1. [Python](#) language – and assorted Python library modules
2. [PostgreSQL](#) database manager

This selection of software is very robust and (except for the DAO drivers) is [Open Source](#).

The selection ensures ongoing zero cost updates indefinitely, and a platform neutral design that can have the PostgreSQL server operating on Windows, MacOS or Linux systems interchangeably, preventing supplier lock-in issues that are common.

After installing the PostgreSQL server, the examples in the document assume you have used the 'psql' utility that gets installed with the server to create a replicator user and passwords.... using a command like this:

```
CREATE USER replicator WITH CREATEDB PASSWORD 'xxxxxxx';
```

Obviously you can use a different username and you definitely should use a better password.

Adjust the replicatorconfig.yaml to match the credentials you select.

You can test the config and your database access with a command like this:

```
C:\Users\ronoh>python replicator.py -c test.yaml --network --tpassword xxxxxxxx
Replication processing starts
2026-05-21 05:41 - INFO - ===== testing network connections =====
2026-05-21 05:41 - INFO - Connected to MS Access database: C:\Users\ronoh\Test.accdb
2026-05-21 05:41 - INFO - MS Access connection: SUCCESS
2026-05-21 05:41 - ERROR - PostgreSQL connection failed: connection to server at "localhost"
(::1), port 5432 failed: FATAL: password authentication failed for user "replicator"

2026-05-21 05:41 - ERROR - Check that:
2026-05-21 05:41 - ERROR - 1. PostgreSQL server is running
2026-05-21 05:41 - ERROR - 2. Host and port are correct
2026-05-21 05:41 - ERROR - 3. Username and password are correct
2026-05-21 05:41 - ERROR - 4. Database exists
2026-05-21 05:41 - ERROR - PostgreSQL connection: FAILED - connection to server at
"localhost" (::1), port 5432 failed: FATAL: password authentication failed for user "replicator"

2026-05-21 05:41 - INFO - ===== network test completed =====
```

Replication processing completed – successful

If you configure or supply the correct password when the PostgreSQL database has not been created (which you could do before any of this, using the psql command line tool)

```
C:\Users\ronoh>python replicator.py -c test.yaml --network --tpassword xxxxxxxx
Replication processing starts
2026-05-21 05:44 - INFO - ===== testing network connections =====
2026-05-21 05:44 - INFO - Connected to MS Access database: C:\Users\ronoh\Test.accdb
2026-05-21 05:44 - INFO - MS Access connection: SUCCESS
2026-05-21 05:44 - ERROR - PostgreSQL connection failed: connection to server at "localhost"
(::1), port 5432 failed: FATAL: database "test" does not exist
```

```
2026-05-21 05:44 - ERROR - Check that:
2026-05-21 05:44 - ERROR - 1. PostgreSQL server is running
2026-05-21 05:44 - ERROR - 2. Host and port are correct
2026-05-21 05:44 - ERROR - 3. Username and password are correct
2026-05-21 05:44 - ERROR - 4. Database exists
2026-05-21 05:44 - ERROR - PostgreSQL connection: FAILED - connection to server at
"localhost" (::1), port 5432 failed: FATAL: database "test" does not exist
```

```
2026-05-21 05:44 - INFO - ===== network test completed =====
```

Replication processing completed – successful

But because the user you created is allowed to create databases, you can have the replicator do this by using the –schema option (remove the –network test option)

```
python replicator.py -c test.yaml --schema --tpassword xxxxxxxx
```

```
Replication processing starts
2026-05-21 05:48 - INFO - Dropped database: test
2026-05-21 05:48 - INFO - Created database: test
2026-05-21 05:48 - INFO - Connected to MS Access database: C:\Users\ronoh\Test.accdb
2026-05-21 05:48 - INFO - Connected to PostgreSQL database: test
2026-05-21 05:48 - INFO - Discovered 4 foreign keys from MS Access (after exclusion filtering)
2026-05-21 05:48 - INFO - Configured 0 foreign keys from YAML
2026-05-21 05:48 - INFO - Total foreign keys: 4
2026-05-21 05:48 - INFO - No 'tables' section in configuration - auto-discovering all non-system
tables
2026-05-21 05:48 - INFO - Auto-discovered 4 non-system tables
2026-05-21 05:48 - INFO - Created table: ecgs
Created table: ecgs
2026-05-21 05:48 - INFO - Created table: events
Created table: events
2026-05-21 05:48 - INFO - Created table: investigations
Created table: investigations
2026-05-21 05:48 - INFO - Created table: patients
```

Created table: patients
 2026-05-21 05:48 - INFO - Created index: patientsevents on events
 2026-05-21 05:48 - INFO - Created index: lab_id on investigations
 2026-05-21 05:48 - INFO - Created index: patientsinvestigations on investigations
 lots more
 2026-05-21 05:48 - INFO - Processing foreign key: fk_patients_investigations
 2026-05-21 05:48 - INFO - Direction analysis for patients(['patient_number']) <->
 investigations(['patient_number'])
 2026-05-21 05:48 - INFO - patients: PRIMARY KEY=True, UNIQUE=False
 2026-05-21 05:48 - INFO - investigations: PRIMARY KEY=False, UNIQUE=False
 2026-05-21 05:48 - INFO - Decision: patients has PRIMARY KEY -> Parent, investigations ->
 Child
 2026-05-21 05:48 - WARNING - Foreign key fk_patients_investigations direction is
 INCORRECT - needs swapping
 2026-05-21 05:48 - INFO - Original: patients(['patient_number']) ->
 investigations(['patient_number'])
 2026-05-21 05:48 - INFO - Corrected: investigations(['patient_number']) ->
 patients(['patient_number'])
 2026-05-21 05:48 - INFO - Processing foreign key: fk_patients_investigations
 2026-05-21 05:48 - INFO - Direction analysis for investigations(['patient_number']) <->
 patients(['patient_number'])
 2026-05-21 05:48 - INFO - investigations: PRIMARY KEY=False, UNIQUE=False
 2026-05-21 05:48 - INFO - patients: PRIMARY KEY=True, UNIQUE=False
 2026-05-21 05:48 - INFO - Decision: patients has PRIMARY KEY -> Parent, investigations ->
 Child
 2026-05-21 05:48 - INFO - Foreign key fk_patients_investigations direction is correct
 2026-05-21 05:48 - INFO - Direction analysis for investigations(['patient_number']) <->
 patients(['patient_number'])
 2026-05-21 05:48 - INFO - investigations: PRIMARY KEY=False, UNIQUE=False
 2026-05-21 05:48 - INFO - patients: PRIMARY KEY=True, UNIQUE=False
 2026-05-21 05:48 - INFO - Decision: patients has PRIMARY KEY -> Parent, investigations ->
 Child
 2026-05-21 05:48 - INFO - Given direction investigations -> patients is VALID
 2026-05-21 05:48 - INFO - Created foreign key: fk_patients_investigations
 2026-05-21 05:48 - INFO - Schema replication only - data copy skipped
 2026-05-21 05:48 - INFO - Connections closed

Replication processing completed - successful

Processes:

Replication and sanitising data

The replicator program by default will replicate all the tables it finds in the source MS Access database into the target PostgreSQL database. If the configuration file specifies table name(s) to replicate it will replicate only those tables.

*Most of the time you should not have a list of table names in the configuration file.. the program will **automatically replicate ALL tables, indexes and foreign key constraints** in the database*

The table names will be the same, the field names will be the same, and data types are mapped as follows

""""Map MS Access data types to PostgreSQL types""""

- "autonumber" or "counter" maps to "SERIAL"
- "long" or "integer" maps to "INTEGER"
- "text", "char" or "memo" maps to "TEXT"
- "date" maps to "DATE"
- "varchar" maps to "VARCHAR"
- "time" or "datetime" maps to "TIMESTAMP"
- "yesno", "boolean" or "bit" maps to "BOOLEAN"
- "currency", "decimal", "double", or "float" maps to "NUMERIC"
- any other datatype maps to "TEXT"

Any datatype not matching any of these (including OLE Object, Hyperlink, Attachment, etc.) maps to "TEXT" as the default case.

Table names, index names and column names may be transformed to all lower case but always in accordance with PostgreSQL rules. As follows:

Unquoted identifiers (safe default): letters (including non-ASCII), digits (0–9) and underscore (_); must start with a letter or underscore; dollar sign (\$) is allowed by PostgreSQL but not SQL standard; unquoted names are folded to lower case; maximum length is 63 bytes by default.

Quoted identifiers (double quotes "like this"): can contain any character except NUL (U+0000); a double quote inside the name is written as two double quotes (""); retain case exactly; still limited to 63 bytes by default.

Notes: SQL keywords may be used only when quoted; identifiers longer than the limit are truncated.

The Primary key of each table in the PostgreSQL database is created to match the Primary key in the MS Access database.

All other keys of each table in the PostgreSQL database are created to match the original keys in the MS Access database.

If foreign key constraints are given in the configuration file they will be applied to the PostgreSQL tables. But if none are configured [the normal setup] they will be discovered from the Access database and those constraints will be applied to the PostgreSQL database tables.

**** Limitations ****

Foreign keys of each table the PostgreSQL database are *automatically* created to match the data. You can define the foreign keys that you want created by entering their details in the configuration file, but this is optional. The default action is to create Foreign key constraints that match the MS Access database.

The replicator applies the PostgreSQL [identifiers](#) rules to table names, index names and column names

By default, the tables created in the target database have

- the same name as the tables source database
- the same column names as the tables have in source database
- the same primary key as the tables have in the source database
- the same foreign key structures as the tables have in the source database

But these names will be translated to lower case unless they need to be “quoted” to conform with the PostgreSQL rules (mixed case names with special characters or spaces)

and read access (SELECT) is set to Anyone ... if you need different access permissions you can change that using PostgreSQL directly or by using a nice GU administrator program like PgAdmin or Webmin

At the moment, by default, the replicator does not replicate deletions in the source MS Access database. This is supported by the command line option `–sync-deleted`, but it implies a second pass that reviews the PostgreSQL tables and looks up each row in the MS Access database, then removes rows in the PostgreSQL database that it can not find in the MS Access database ...

A second pass is slower than simply running the replicator with the `–full-refresh` command line option, which achieves the same result – though it makes the PostgreSQL database incomplete while it runs, until the run completes.

See the `–sync-deleted` section of this document

MS Access Relationships are not quite Foreign keys

During testing on a sample database from the [Kaggle](#) community website, I encountered some odd messages from the fairly defensive code in the replicator.

2026-05-21 01:56 - INFO - Creating foreign keys...

2026-05-21 01:56 - INFO - Processing foreign key: fk_dimtarih_factsatis

2026-05-21 01:56 - INFO - Direction analysis for DimTarih(['TarihKod']) <-> FactSatis(['SatisTarihi'])

2026-05-21 01:56 - INFO - DimTarih: PRIMARY KEY=False, UNIQUE=False

2026-05-21 01:56 - INFO - FactSatis: PRIMARY KEY=False, UNIQUE=False

2026-05-21 01:56 - WARNING - Decision: Neither table has PRIMARY KEY or UNIQUE constraint

2026-05-21 01:56 - INFO - Direction analysis for DimTarih(['TarihKod']) <-> FactSatis(['SatisTarihi'])

2026-05-21 01:56 - INFO - DimTarih: PRIMARY KEY=False, UNIQUE=False

2026-05-21 01:56 - INFO - FactSatis: PRIMARY KEY=False, UNIQUE=False

2026-05-21 01:56 - WARNING - Decision: Neither table has PRIMARY KEY or UNIQUE constraint

2026-05-21 01:56 - WARNING - Foreign key 'fk_dimtarih_factsatis' cannot be created: neither table has PRIMARY KEY or UNIQUE constraint

So I asked DeepSeek: In MS Access you can define relationships between tables, but these do not seem to match SQL foreign keys because they don't seem to need indexes on the tables

The answer was interesting – shown in the next section,

BUT not having the useful indexes is simple to resolve in Access itself -- just add a primary key to each table and other needed indexes ... the replicator runs fine even if you don't do that, but may not create foreign keys which you might like to have.

This was previous just a documented limitation .. with a simple correction possible by using Access to adjust the table definition(s). Add Primary keys or unique keys that are needed in relationships defined to your Access database.

In the current version, the replicator will attempt to create enough primary keys or unique indexes to satisfy implementing the MS Access relationships as PostgreSQL foreign keys.



MS Access relationships vs. SQL foreign keys

Why Access table relationships don't *require* indexes — and how that differs from true FK constraints

Your sharp observation: “In MS Access you can define relationships between tables, but these do not seem to match SQL foreign keys because they don't seem to need indexes on the tables.”

You are absolutely right: a basic Access relationship (just a line) is **not** the same as a SQL foreign key. It acts more like a **query design hint** rather than an integrity rule. Indexes become optional depending on whether you enforce referential integrity.

MS Access relationship (simple link)

When you drag a line between two fields in the Relationships window **without** checking **Enforce Referential Integrity**:

- It's purely **informational** / **visual** for the query designer.
- Access will **auto-join** tables in queries, but no physical rule is stored.
- Orphan records **are allowed** (you can add an order with a missing CustomerID).
- No index required** on the foreign-key side — Access can use hash or loop joins in memory.



Design-time only · No ON DELETE CASCADE · No automatic index creation



key point Without enforcement, the "relationship" is just metadata — it doesn't behave like an FK constraint.



SQL foreign key (true constraint)

In PostgreSQL, MySQL, SQL Server, or Oracle, a foreign key creates a **physical integrity rule**:

- Prevents orphan rows (child records must reference an existing parent).
- Supports **ON DELETE CASCADE / SET NULL**.
- Requires an index** on the foreign key column to maintain referential integrity efficiently.
- The index is often created automatically by major DBMS (or mandatory for performance).

```
ALTER TABLE orders ADD CONSTRAINT fk_customer
```

```
FOREIGN KEY (customer_id) REFERENCES customers(id);
```



critical Without an index, FK checks would cause full table scans, so SQL engines strongly require/auto-create indexes.



The missing piece: “Enforce Referential Integrity” changes everything

If you double-click the relationship line in Access and check “**Enforce Referential Integrity**”, Access *behaves almost exactly like a SQL foreign key*. At that moment:

- ❌ You can't create orphan records (child must have a matching parent).
- ❌ You can't delete a parent record if child records exist (unless you cascade).
- 📈 **Index requirement:** Access will either prompt you to create an index or **implicitly creates a hidden index** on the foreign key column. You just don't see it in the table design grid.

👉 So why doesn't a simple (non-enforced) relationship need an index? Because it doesn't enforce anything — it's just a visual & query auto-join hint. No constraint = no need for the engine to constantly verify child rows.

📊 Feature comparison table

Prevents orphan records	❌ No	✅ Yes	✅ Yes
Requires / uses index on FK side	❌ Not required (optional for query speed)	✅ Yes (implicit or explicit index required)	✅ Required / auto-created
Can you see the index in table design?	No, unless manually added	Sometimes hidden, but engine relies on it	Visible in schema (index on FK column)
Query auto-join behaviour	✅ Yes (designer hint)	✅ Yes	Not relevant (physical constraint)
ON DELETE CASCADE option	No	✅ Optional	✅ Optional
Performance overhead for inserts/updates	Minimal (no check)	Moderate (referential check & index update)	Moderate–High (index + constraint verification)

⚡ Why doesn't a basic Access relationship require indexes?

Because the Jet/ACE database engine is file-based, not server-based. For modest row counts (< 100k–200k), Access can perform **hash joins or nested loop joins in memory** faster than it can traverse a disk-based index. But the moment you *Enforce Referential Integrity* or need high performance on millions of rows, you **must** add an index. The rule is: *no constraint = no forced index; constraint = index needed*. That's why simple relationships feel different from SQL foreign keys.

🔧 Practical demonstration (Mental model)

🔗 Non-enforced relationship

- ✅ No index required
- ✅ Allows orphan records
- ⚠ Joins might be slower on large data
- ✅ **Enforced RI (acts like FK)**
- 🔍 Hides/creates index behind the scenes

❌ Rejects invalid foreign keys

🧠 Closest to SQL foreign key

⚙ SQL foreign key

- 📋 Mandatory index (explicit)
- 🔒 ACID + constraint strictness
- 🚀 Designed for large workloads



When are indexes actually required in Access relationships?

- **Simple line (no RI):** Index is optional. Helps join performance on large tables but not forced.
- **Enforced RI:** Access will either **demand an index** or create one implicitly. If you try to enforce RI without an index, Access often shows the warning: “Do you want to create an index on these fields?”
- **Join queries:** Even without relationships, an index on foreign key column drastically improves speed.



Therefore, your original statement holds true: *basic relationships do not 'match' SQL foreign keys because they lack forced referential integrity and mandatory indexes*. The moment you enable enforcement, Access transforms the relationship into something functionally equivalent — including hidden reliance on indexes.



Final answer – why the mismatch feels confusing

Many data professionals assume that drawing a relationship line in Access creates a **full foreign key constraint** like in SQL. It does not — unless you explicitly **Enforce Referential Integrity**. Without that checkbox, Access treats the relationship as a lightweight instruction for query design, not a database integrity rule, which is why indexes aren't mandatory. That's the root of the difference you observed.



Bottom line: For strict data integrity + index-backed foreign key behavior in MS Access, always check “Enforce Referential Integrity” and verify that an index exists (or let Access create it). Otherwise, you just have a visual guide – not a real foreign key.



Insight based on Access Jet/ACE engine & standard SQL behavior · your question highlights one of the most common misconceptions among SQL users transitioning to Access.

Handling MS Access Relationships During PostgreSQL Migration

When migrating a Microsoft Access database to PostgreSQL, a common challenge arises from the fundamental differences in how each system enforces data integrity. Microsoft Access allows you to define relationships in its Relationship Window, even if the underlying tables do not have properly indexed columns to support a formal foreign key constraint in a SQL database.

This document describes the program's behavior and technical requirements when converting Access relationships into PostgreSQL foreign keys, using standard SQL parent/child terminology. The "parent" table is the primary table in a relationship (the "one" side), and the "child" table is the referencing table (the "many" side).

This behaviour can be suppressed by using the `--no-auto-index` command line option. If that option is used you will see messages about any table that is affected ... where the equivalent foreign key was not created and the MS Access relationship has not been replicated.

The Core Challenge: Indexing

In PostgreSQL (and most relational database management systems), a foreign key constraint, which defines a parent/child relationship, requires a unique index on the referenced columns of the parent table. This is a strict technical requirement.

By contrast, Microsoft Access is more flexible. You can create a relationship in the Relationship Window between a parent and a child table, even if the parent table's primary key field lacks a unique index. For example, if the parent table's key field allows duplicate values, Access may still accept the relationship definition, but its enforcement is less rigorous.

Program Behaviour During Migration

When the program analyses a parent/child relationship from MS Access, it performs a specific validation check to determine if it can be safely converted into a PostgreSQL foreign key.

Analysis of the Parent Table: The program identifies the column in the parent table that the relationship is based upon.

Index Validation: It checks if this parent column has a unique index.

Decision Point:

- If a unique index exists: The program proceeds. It will create the corresponding foreign key constraint in PostgreSQL using the standard ALTER TABLE command. This ensures data integrity is maintained at the database level.
- If a unique index does NOT exist: The program cannot create a standard foreign key, as this would violate PostgreSQL's core rules.

Handling Unsuitable Parent Keys

When the parent table lacks a unique index, the program employs a strategy to avoid errors and preserve the logical link.

The Relationship is Skipped: The program will not create a formal FOREIGN KEY constraint in the PostgreSQL schema. This prevents a migration failure that would otherwise occur if it attempted to create an invalid constraint.

Integrity Enforcement is Shifted: The responsibility for maintaining the parent/child relationship is not dropped, but it is shifted from the database to the application layer.

- The PostgreSQL Database: Will manage the two tables as independent entities. It will not automatically check that every value in the child's foreign key column has a matching value in the parent's referenced column.
- The Application Layer: The program recommends or relies on the application using the database to enforce the relationship logic. This is often done through application code or, in the case of a front-end like a migrated Access project, through queries and business logic rules.

Summary – Foreign key issues

The program prioritizes a successful and stable migration. It enforces the strict requirement for a unique parent index. If this requirement is not met, it safely skips creating the formal constraint, leaving the relational logic to be handled by the application to prevent a migration failure.

Sample console messages for --no-auto-index

2026-05-21 23:30 - INFO - Processing foreign key: fk_dimaltkategori_dimurun
2026-05-21 23:30 - INFO - Direction analysis for DimAltKategori(['AltKategoriKod']) <->
DimUrun(['AltKategoriKod'])
2026-05-21 23:30 - INFO - DimAltKategori: PRIMARY KEY=False, UNIQUE=False
2026-05-21 23:30 - INFO - DimUrun: PRIMARY KEY=False, UNIQUE=False
2026-05-21 23:30 - WARNING - Decision: Neither table has PRIMARY KEY or UNIQUE constraint
2026-05-21 23:30 - INFO - Direction analysis for DimAltKategori(['AltKategoriKod']) <->
DimUrun(['AltKategoriKod'])
2026-05-21 23:30 - INFO - DimAltKategori: PRIMARY KEY=False, UNIQUE=False
2026-05-21 23:30 - INFO - DimUrun: PRIMARY KEY=False, UNIQUE=False
2026-05-21 23:30 - WARNING - Decision: Neither table has PRIMARY KEY or UNIQUE constraint

=====

 FOREIGN KEY SKIPPED (--no-auto-index enabled):

Foreign key name: fk_dimaltkategori_dimurun

Child table (base_table) : DimAltKategori

Parent table (reference) : DimUrun

Child columns : AltKategoriKod

Parent columns : AltKategoriKod

This foreign key requires a PRIMARY KEY or UNIQUE constraint on the child
table columns, but automatic creation is disabled.

The foreign key will NOT be created.

=====

**** data sanitising / transforming ****

You may wish to sanitise the data before it is replicated into the PostgreSQL data system, individual data items may be identified as not to be replicated, or edited according to a transformation rule before being replicated

During replication, existing rows in the target table(s) are updated to match any revised data.

New rows that are not yet in the replicated target system are simply inserted after transformation rules are applied.

Transformation of fields is based on configuration file entries that specify individual field names in a table that should have their data altered to preserve the security of the Clinical Practice data,

The AI research can ofent function properly even if some of the data is obscured

Note: having new fields specified as 'Drop' will not clear any data from any prior runs of the replicator. In this case, you should use the `–full-refresh` command line option to force previously copied data to be removed. After that you can resume normal operation for incremental updates.

The transformations possible are:

[MMH3](#) replace a field with the MurmurHash3 hash of it – normal used to obscure any integer

YearOnly replace a full date with 1st of January of the same year

Drop Don't copy this field

An example of the configuration file entries

```
# Transformations - MUST be a LIST of dictionaries
```

```
transformations:
```

```
- table: "Dimindirim"
```

```
columns:
```

```
- name: "Baslangic"
```

```
transform: "YearOnly"
```

Command line switches and their usage

There are a number of situations where you may want to use the optional command line switches, and this part of the document tries to guide when each switch could be used.

Minimum configuration file

The basic use of the replicator is to have a configuration file (replicatorconfig.yaml) with the connection information needed for the Access input database, and the PostgreSQL target database.

DAO:

```
driver: "Microsoft Access Driver (*.mdb, *.accdb)"
```

```
database: "ContosoTR.accdb"
```

```
# Target PostgreSQL connection
```

```
postgresql:
```

```
host: "localhost"
```

```
port: 5432
```

```
database: "contosotr"
```

```
user: "replicator"
```

```
password: "xxxxxxx"
```

```
# Global settings
```

```
global:
```

```
debug: false
```

```
managerid: "Replicator"
```

```
verbose: true
```

Then the command could be: **python replicator.py**

If the configuration file does not have a *tables:* entry the default action is to create a complete copy of the tables found in the Access database, into the PostgreSQL database. A compatible schema for each Access table with columns, indexes and foreign key constraints is created in the PostgreSQL database (if not already created), and then the data for every table is copied or updated (with optional transformations) into the PostgreSQL database.

You can use the same configuration file without a *tables:* entry with the **-list** command line option to get exactly that. A list of the table names in your database.

MS Access tables without keys

The replicator can and will copy tables without keys – append only. If the PostgreSQL database is empty or you run it with the `--full-refresh` command line option this is fine.

But if you re-run it normally, the tables with keys get updated, but the tables without keys get all the data appended to the output table **again..** almost certainly not what you would like.

There is a command line option to correct this situation but ****WARNING**** **it changes the MS Access database structure.** This is the only replicator function that writes to your input MS Access database so you should take a backup before you start.

It is also the only command line option that does not require valid PostgreSQL configuration data, So it can operate independently on the MS Access database. ****WARNING**** since it adds an AutoNumber field (if needed) to your tables, and if the tables have a large number of rows this will be a very slow, one time operation. If a table has an AutoNumber field, but no Primary key or unique key, the replicator will not add anew column. It will change the Access database to use that existing AutoNumber field as a primary key. That field **should** be valid as a primary key – but MS Access has been known to have duplicate values in an AutoNumber field ... and there will be an error message given if that happens. Your data will need to be corrected manually to fix this. A simple fix would be to change that column to Long Integer rather than AutoNumber – but that could cause problems with your Access application.

The option is called `--adjust-ms-access`

Running the replicator with this option is like having a stand alone MS Access table editing function.

--adjust-ms-access

DESCRIPTION

This option modifies the schema of an MS Access database by adding an AutoNumber primary key column to every user table that does not already have a primary key.

The operation is performed directly on the Access database file specified in the configuration file or via the --source parameter. No connection to PostgreSQL is required or attempted when this option is used.

PURPOSE

Many MS Access tables lack explicit primary keys, which can cause issues when replicating data to PostgreSQL or when attempting to establish relationships. This function ensures that every user table has a unique, auto-incrementing identifier column that can serve as a primary key.

PROCESSING DETAILS

1. The program connects to the MS Access database using DAO.
2. It iterates through all non-system user tables.
3. For each table without a primary key:
 - A new column is added with the name: {table_name}_replicator_id
 - The column is created as an AutoNumber (Long Integer) type, which automatically generates sequential numbers for existing rows.
 - A PRIMARY KEY constraint is created on this new column.
 - The column name is sanitised (lowercase, spaces replaced with underscores, non-alphanumeric characters removed) and truncated to 64 characters if necessary.
4. Tables that already have a primary key are skipped.
5. Tables that already contain a column with the same name are skipped.

6. System tables (names starting with MSys or ~) are excluded from processing.

OUTPUT

For each table processed, the program displays:

-  Table name: already has PRIMARY KEY - skipping
-  Table name: adding AutoNumber primary key column 'column_name'
✓ Successfully added primary key on 'column_name'
-  Table name: column 'column_name' already exists - skipping
-  ERROR on table name: (error description)

At the end of processing, a summary is displayed showing:

- Tables modified
- Tables skipped
- Errors encountered

EXIT CODES

The program exits with code 0 on successful completion.

Exits with non-zero code if the Access database cannot be opened or if a fatal error occurs.

USAGE EXAMPLES

Basic usage with configuration file

```
python replicator.py --adjust-ms-access
```

Specify configuration file and source database

```
python replicator.py --adjust-ms-access -c myconfig.yaml -s C:\data\database.accdb
```

Verbose output

python replicator.py --adjust-ms-access -v

NOTES

- This option is mutually exclusive with --schema, --list, --network, --dump, and --full-refresh.
- PostgreSQL connection parameters are ignored when this option is used.
- The Access database must not be open exclusively by another user.
- Existing rows automatically receive sequential numbers when the AutoNumber column is added.
- The new primary key is named "PrimaryKey" in Access (default naming).
- This operation modifies the source Access database. It is recommended to backup your database before running this command.

RELATED OPTIONS

- source, -s Specify the Access database file path
- config, -c Specify the configuration file (default: replicatorconfig.yaml)
- verbose, -v Display detailed progress information

TYPICAL OUTPUT LOG

```
2026-05-22 02:03 - INFO - Connected to MS Access database: C:\Users\ronoh\Test.accdb
2026-05-22 02:03 - INFO - Connected to PostgreSQL database: test
2026-05-22 02:03 - INFO - Connected to MS Access database: C:\Users\ronoh\Test.accdb
2026-05-22 02:03 - INFO - adjust_ms_access_schema: DAO connection successful
2026-05-22 02:03 - INFO - adjust_ms_access_schema: Found 5 user tables
2026-05-22 02:03 - INFO - adjust_ms_access_schema: ecgs already has PK, skipping
2026-05-22 02:03 - INFO - adjust_ms_access_schema: events already has PK, skipping
2026-05-22 02:03 - INFO - adjust_ms_access_schema: investigations already has PK, skipping
2026-05-22 02:03 - INFO - adjust_ms_access_schema: patients already has PK, skipping
2026-05-22 02:03 - INFO - adjust_ms_access_schema: Adding PK to some-log with column
some_log_replicator_id
2026-05-22 02:03 - INFO - adjust_ms_access_schema: Successfully added PK to some-log
```


2026-05-22 02:03 - INFO - Connections closed

2026-05-22 02:03 - INFO - adjust_ms_access_schema completed: 1 modified, 4 skipped, 0 errors

~

~

--sync-deleted

Purpose

The --sync-deleted option synchronizes record deletions from the source Microsoft Access database to the target PostgreSQL database. During normal replication, the program copies new and updated records from Access to PostgreSQL but does not remove records that have been deleted from the source. This option addresses that gap by identifying and removing records from PostgreSQL that no longer exist in Access.

How It Works

When --sync-deleted is enabled, the program performs the following operations after the main data copy phase is complete:

1. Row Count Comparison – For each table being replicated, the program compares the total number of rows in the Access source against the row count in the PostgreSQL target.
2. Selective Processing – By default (fast mode), only tables with differing row counts are processed. Tables with matching row counts are skipped, as no deletions are needed.
3. Key-Based Row Identification – For each table requiring synchronization, the program identifies which rows in PostgreSQL should be deleted by:
 - Using the table's PRIMARY KEY if one exists
 - Falling back to the first available UNIQUE constraint if no PRIMARY KEY is defined
 - If neither exists, the table is skipped with a warning (deletion synchronization requires a way to uniquely identify each row)
4. Batch Processing with Keyset Pagination – To avoid memory issues with large tables, the program processes rows in batches of 1,000 records. For each batch:
 - Rows are retrieved from PostgreSQL in sorted order by the key columns
 - Each row's key value is checked against the Access database using a SQL COUNT query
 - If the key value is not found in Access, the corresponding PostgreSQL row is marked for deletion

- Marked rows are deleted from PostgreSQL using a batched DELETE statement

5. Progress Reporting – When --verbose is enabled, the program displays:

- A progress bar showing completion percentage
- Estimated time remaining (ETA) for the current table
- Number of rows deleted so far
- Final count of deleted rows for each processed table

6. Validation – After deletion synchronization completes, the program re-validates row counts and displays a summary showing which tables were processed and how many rows were deleted.

The --slow Option

When --slow is specified together with --sync-deleted, the program changes its behaviour in the following way:

Mode	Behaviour
Fast Mode (default)	Only tables with differing row counts between Access and PostgreSQL are processed
Slow Mode (--slow)	Every table is processed regardless of whether row counts match

When to use Slow Mode:

Slow mode is useful in the following scenarios:

- Silent data corruption – When row counts match but individual records may have been deleted and reinserted with different key values, leaving orphaned records with different identifiers
- Foreign key cascade testing – When verifying that cascade deletions are properly propagating through related tables
- Recovery from interrupted syncs – When a previous sync-deleted operation was interrupted and the row counts no longer accurately reflect the state of deletions

- Debugging and validation – When performing thorough integrity checks on the synchronization process

Performance Note: Slow mode significantly increases processing time because every table is fully scanned regardless of whether deletions are needed. It should only be used when the additional thoroughness is required.

Important Requirements

For --sync-deleted to function correctly:

- Each table must have a PRIMARY KEY or at least one UNIQUE constraint defined in PostgreSQL
- Foreign key relationships should be defined with ON DELETE CASCADE where appropriate to maintain referential integrity
- The Access and PostgreSQL databases must be otherwise synchronized (run a normal replication first)

Example Usage

Standard deletion synchronization (fast mode):

```
python replicator_v1.34.py --config myconfig.yaml --sync-deleted --verbose
```

Slow mode (process all tables):

```
python replicator_v1.34.py --config myconfig.yaml --sync-deleted --slow --verbose
```

Typical Output

```
=====
SYNC DELETED RECORDS
=====
```

Fast mode - will only process tables where row counts differ.

Other ways to use the minimum configuration file

You can also use the same configuration file without a *tables:* entry with the **–output {filename}** command line option to generate a different yaml format configuration file with all the discovered schema structure in listed in the new configuration file. This is handy if you want to move some table names that were discovered into the **excluded:** section to suppress the replication of them

If run the replication with a minimum configuration file that specifies an Access database, (and does not have a *tables* section:) with the **–output {filename}** command line option . It will generate a yaml format configuration file with all the discovered schema structure in listed in the new configuration file. That will include the definitions of the foreign keys .. this new yaml configuration file can be edited - usually to move some table names into the *excluded* section

--verbose – puts more messages on the console and in the log. If you are watching the process you probably want **--verbose** so that you get a progress indicator and a completion time estimate as each table is processed.

--schema – **Deletes the existing PostgreSQL** database and recreates the tables in it, but does not replicate the data. Useful if you wish to start again from a clean position

--full-refresh ... the same as using **--schema** and then running it again to replicate the data

--no-auto-index suppresses the program behaviour needed to replicate MS Access relationships

--sync-deleted runs a second pass to remove rows from the PostgreSQL tables that have been deleted in the MS Access tables. As an optimisation it assumes that if the number of rows in table is the same in both databases, then that table does not need to be scanned (true unless you have deleted and inserted the same number of rows)

--slow runs the **--sync-deleted** logic on all tables even if the row counts for the source and target are the same.

The following command line options are only for developer use

--debug

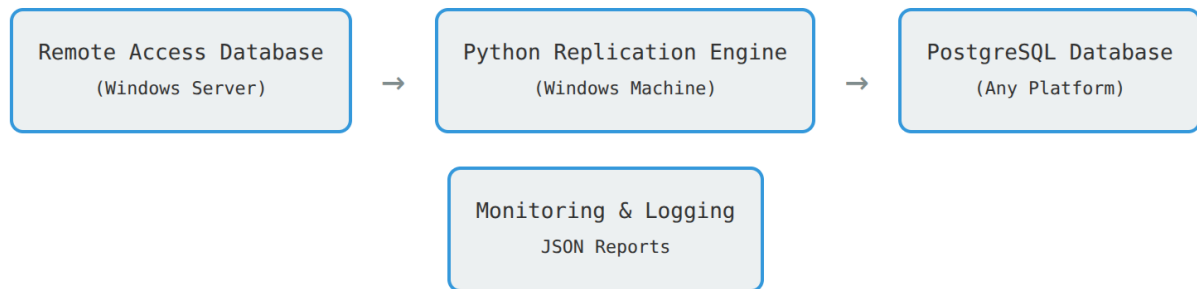
--trace

--dump

Appendix 1 – Software for the replicator

The replicator tool runs in a Windows machine and is a Python program that uses [DAO](#) driver to let it read the MS Access database that holds your data.

The program also connects to a PostgreSQL database server running in the target machine (either MacOS , Linux or Windows Pro) using the Python library module for PostgreSQL client access.



Software required in the machine running the replicator

Mandatory

1. If Access is not installed, DAO/ODBC/Jet driver for MS Access: [ODBC download](#) (free)
2. Python 3.x client library – the full server is not required (free)
3. modules installed using ‘pip’ `execsql pywin32 pycpg2-binary mmh3 schedule dotenv pyyaml pycpg2-binary pandas` (free)
4. the replicator program - `replicator.py` (free)
5. replicator configuration file – `replicatorconfig.yaml` (adjust to suit DAO database name, PostgreSQL server IP address, database name and credentials)

Recommended

6. UniGetUI 3.3.6 (from GitHub) - global package manager (free)

The replicator program checks its configuration file and decides which type of connection to use depending on which format of parameters is present in the file. The configuration file also holds the details of the PostgreSQL server and the access credentials needed. The default configuration file name is “**replicatorconfig.yaml**”

```
# Source DAO connection
DAO:
  driver: "Microsoft Access Driver (*.mdb, *.accdb)"
  database: "ContosoTR.accdb"

# Target PostgreSQL connection
postgresql:
  host: "localhost"
  port: 5432
  database: "contosotr"
  user: "replicator"
  password: "xxxxxx"

# Global settings
global:
  debug: false
  managerid: "Replicator"
  verbose: true

# Tables to replicate (commented out - will auto-discover from database)
#tables:
# - "DimKanal"
# - "FactSatis"

excluded:
  - "Events"

# Primary keys for tables (optional - will auto-discover)
# primarykeys:
#   xxxxs:
#     - "customer_number"
#   Visits:
#     - "VisitID"

# Transformations - MUST be a LIST of dictionaries
transformations:
  - table: "Dimindirim"
    columns:
      - name: "Baslangic"
        transform: "YearOnly"

# Foreign keys (optional - will auto-discover)
# foreignkeys:
#   fk_patients_investigations:
#     base_table: "DimCheck"
#     reference_table: "Dimindirim"
#     base_columns:
#       - "Urunkod"
#     reference_columns:
#       - "Urunkod"
```

Transformations are of course optional, and the production settings of the desired transformations will need discussion about the actual data privacy needs.

You can obviously have multiple different configuration files. This allows the replication to be targeted at multiple research systems if needed..... just run the replicator once for each target system that needs to be updated, and specify the configuration file on the command line like this:

```
python replicator.py --config secondconfig.yaml
```

This the default behaviour of this replicator program is that it is a generic MS Access database to PostgreSQL database converter. It is not tied to the specific data structures of the Clinical AI project.

Normally processing messages and error messages are displayed in the command window during execution.

In addition most messages are also logged to a file called 'replicator.log' .The log file is automatically rotated when it exceeds 10 megabytes in size. A limit of 10 generations of historical log files is retained. The historical log files have a ".1". "2" or similar suffix appended to the file name.

Installing the replicator

There are 5 files that need to be placed in the users home directory on the Windows system that holds the Access database to be replicated.

- replicator.py
- run_replicator_normal.bat
- run_replicator_refresh.bat
- run_replicator_help.bat
- replicatorconfig.yaml

For ease of operation you will want to create a New->Shortcut on your desktop for each of the .bat files.

Install python ([Video instructions](#))

In addition to installing python you need also open a CMD window and issue the following command to install the extra python modules used by the replicator.py code

```
pip install execsql pywin32 psycopyg2-binary mmh3 schedule dotenv pyyaml psycopyg2-binary pandas
```

**** configuration file notes ****

The configuration file is what you normally use to control the replicator, and for most uses the only thing it needs to contain, is the Access database file name, the access credentials and and [PostgreSQL](#) connection details. The 'global' section can also be set up if you like.

You would normally only use the command line options for things the configuration file does not support (like --schema, --debug, --trace)

The replicator copies (or updates) and transforms the tables without foreign keys first before copying tables with foreign keys that depend of data already being held in another table.

replicator command line usage

python replicator.py --help

```
C:\Users\ronoh>python replicator.py --help
usage: replicator.py [-h] [-V] [-c CONFIG] [-s SOURCE] [--thost THOST]
                    [--tport TPORT] [--tdatabase TDATABASE] [--tuser TUSER]
                    [--tpassword TPASSWORD] [-v] [-o OUTPUT] [--debug] [--trace]
                    [-a] [--sync-deleted] [--slow] [-S | --adjust-ms-access] [-l]
                    [-n | --dump | --full-refresh]

MS Access to PostgreSQL Replication Tool

options:
```

-h, --help	show this help message and exit
-V, --version	Show version and exit
-c, --config CONFIG	Path to configuration file
-s, --source SOURCE	MS Access database file name
--thost THOST	PostgreSQL server host name or IP
--tport TPORT	PostgreSQL server port number
--tdatabase TDATABASE	PostgreSQL database name
--tuser TUSER	PostgreSQL user name
--tpassword TPASSWORD	PostgreSQL password
-v, --verbose	Print informational messages
-o, --output OUTPUT	Output file for generated YAML configuration
--debug	Enable SQL debugging output
--trace	Enable trace logging to file
-a, --no-auto-index	Suppress automatic creation of indexes/constraints for foreign keys (skip FK if needed)
--sync-deleted	Synchronize deleted records from Access to PostgreSQL
--slow	Use slower but safer deletion method (only valid with --sync-deleted)
-S, --schema	Drop and recreate database, then replicate schema ONLY
--adjust-ms-access	Adjust MS Access schema (add AutoNumber primary key to tables without PK)
-l, --list	List table names and exit
-n, --network	Test both source and target connections
--dump	Dump internal program data
--full-refresh	Perform full refresh (drop and recreate all tables)

Appendix 3 – Development notes

Development and initial testing is being done within a fairly powerful HP Spectre notebook which is running Linux Mint. The Clinical Practice test environment is a Windows 11 Pro system installed and running as a [Virtual Machine](#) created and operated using the [Oracle Virtual Box](#) software as a Type 2 [Hypervisor](#)

The Linux Mint computer can run the same (identical) Clinical Research software that would operate in a MacOS environment, so it is a suitable development environment.

There is also a Clinical Research Windows 11 Pro environment available as a [Virtual Machine](#) which in addition to having a full [PostgreSQL server](#) installed has [Access 2024](#) installed to allow creation of test data.

All the code development is done in the Linux machine, and is made available to the Windows VM systems (either of them) as a shared drive Z: using the capabilities of the [Virtual Box](#) software.

Very simple version control of the development output is done using the [RCS](#) version control software.

Project output is created using:

- LibreOffice Writer – documents

- LibreOffice Draw - diagrams

- Spectacle – Linux screen shots

- Kate editor – internal notes and work log

- Vi editor – program editing

- [Greenshot](#) – Windows screen shots

Evolution of the development

During development it became clear that the existing input databases can be the source of a number of problems.

Support for the .mdb database format has been progressively downgraded by Microsoft.

- Access 2024 no longer allows you to create .mdb databases
- Access 2024 no longer allows you to import .mdb database

Other issues.

- one table I used for testing had over 109,000 rows ... but with the .mdb file already over 1Gb in size, the “select *” ran out of space when trying to copy it. Attempts to process 1 row at a time via ODBC which should not create any temporary table as a result set, just do not work.

- and that database is locked into presenting a form at startup for the application users ... but you can compact it by running MSAccess.exe from the command line with the /compact switch at the end of the command. That reduced the size of the .mdb file to 135,020,544 bytes ... clearly some housekeeping was in order. The MSAccess.exe can be installed in a variety of places, but from a CMD prompt to locate the MSAccess executable

```
cd C:\
```

```
dir /s *access*.exe
```

On Windows 11 Pro my Access 2024 command to compact the database was:

```
C:\Program Files\Microsoft Office\root\Office16\MSAccess.exe" "C:\Users\ronoh\xxxxxx.mdb"  
/compact
```

The ‘huge table’ problem encountered while using the ODBC driver as the basis for the replicator was resolved this way.

However, after revising the replicator to use the DAO library rather than ODBC, the ‘huge table’ problem no longer exists and the replicator works properly against the original database.

Test data

A really small test database

Available with the replicator is a very small MSAccess database called “Test.accdb” and a replicator configuration file called “test.yaml” that assumes you are running the PostgreSQL under Windows in the same machine. You need to edit that to set the credentials (user and password) to the user you set up for this process. Any PostgreSQL user with the ‘createdb’ authorisation will do.

You can start verifying the installation by working in a CMD window .. in a the directory with the ‘replicator.py’ program, the ‘Test.accdb’ file and the ‘test.yaml’ configuration file.

#1 check the MS Access database is accessible

```
python replicator.py -c test.yaml --list
```

#2 check that your network connection to PostgreSQL is ok

```
python replicator.py -c test.yaml --network
```

Then you can get the replication done

#3 convert the MS Access database schema to PostgreSQL

```
python replicator.py -c test.yaml --schema
```

#4 replicate the MS Access data to PostgreSQL

```
python replicator.py -c test.yaml --verbose
```

You should review the ‘replicator.log’ to see how well your data was replicated. There may be problems reported. Especially for tables that have MS relationships without suitable indexes and with ‘enforce referential integrity’

Access tables without any keys (primary key OR unique key) do not replicate properly if you are able to revise your MS Access database to correct this please do so .. only a DB analyst can properly assess the impact of any changes made. The replicator has an option --adjust-ms-access which can make adjustments to the Access schema, but it can not assess the potential negative performance impacts.

Other test databases.

The website [Kaggle](#) has a huge library of dataset that can be useful for testing

One particular MS Access dataset that was helpful in testing the replicator was the [ContosoTR](#) data.

It is a very *large* database, with table sizes well in excess of the commonly recommended maximum of 100,000 rows.

This is the config file for that database: (replicator-ConosoTR.yaml)

```
# replicatorconfig.yaml
# Configuration for ODBC to PostgreSQL Database Replication System

# Source DAO connection
DAO:
  driver: "Microsoft Access Driver (*.mdb, *.accdb)"
  database: "ContosoTR.accdb"

# Target PostgreSQL connection
postgresql:
  host: "localhost"
  port: 5432
  database: "contosotr"
  user: "replicator"
  password: "badpassword"

# Global settings
global:
  debug: false
  managerid: "Replicator"
  verbose: true

# Tables to replicate (commented out - will auto-discover from database)
#tables:
#  - "DimKanal"
#  - "FactSatis"

excluded:
  - "Events"

# Primary keys for tables (optional - will auto-discover)
# primarykeys:
```

```
# xxxxs:

# - "customer_number"

# Visits:

# - "VisitID"


# Transformations - MUST be a LIST of dictionaries
transformations:

- table: "Dimindirim"

  columns:

    - name: "Baslangic"

      transform: "YearOnly"


# Foreign keys (optional - will auto-discover)
# foreignkeys:

# fk_patients_investigations:

#   base_table: "DimCheck"

#   reference_table: "Dimindirim"

#   base_columns:

#     - "Urunkod"

#   reference_columns:

#     - "Urunkod"
```